Zhengyang Yu

Software Engineering Student

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium [App Store] [GitHub]**, *Personal Project*
- Engineered a performance-conscious C++/FreeRTOS system with concurrent network workers, adaptive HTTPS polling, framebuffer rendering, IMU-driven physics, and bounded persistent logs on an ESP32-S3.
- Designed for predictable resource use and failure recovery through cached state, rate-limit-aware scheduling, secure device authorization, structured diagnostics, and repeatable physical-device testing.
- Shipped a native SwiftUI companion for iPhone and Apple Watch to the App Store, extending the system with cross-device synchronization, background refresh, and a WidgetKit complication.

2025 **HTML to ePub Converter**, *Personal Project*
- Built a Python transformation pipeline that parses HTML and XML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved correctness and maintainability through modular transformations, explicit error handling, and repeatable output validation.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST workflow with explicit state transitions to support testing, troubleshooting, and reliable smart-lighting behavior.

Skills

Programming C++, Python, Java, Swift, C

Systems Concurrency, FreeRTOS, Linux, networking, REST APIs, HTTPS, caching, structured logging

Engineering Algorithms, data structures, object-oriented design, design patterns, unit and integration testing, debugging

Product Delivery Requirements analysis, technical documentation, App Store release, physical-device validation

AI Development Codex-assisted codebase exploration, implementation, refactoring, debugging, and verification

Tools Git, Docker, Linux, PlatformIO, Xcode, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.